

NFC Book IV Universal Source L4 Shell

Schema v0.40.0

L4 Spine Release Candidate

Completes the full branch-to-source dependency chain

Generated by NFC Governance Engine v40

Contents

Book IV L4 Shell

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Book IV completes the chain. **U** exists (Thm. IV.5.1), is unique (by factorization role), and every lawful realization factors through it. The full spine dependency chain from any branch endpoint back to the source is now machine-traceable in the governance engine. This is an L4 spine-pilot shell — the canonical Book IV is the authoritative source.

Book IV Universal Source — L4 Generated Shell

*Book IV establishes the universal source object $\mathbf{U} \in \text{POT}$: it exists (Thm. IV.5.1), is unique (by factorization role), and every lawful realization factors through it (*cor:IV.descendants-factor*). This pilot completes the full spine dependency chain.*

Five ORA Rungs Building U

Machine ID	St.	Canonical label
thm:IV.ORA	[U]	ORA — Overlap Route-Agreement Theorem [U]
thm:IV.CTM	[U]	CTM — Coherent Transition-Map Theorem [U]
thm:IV.normalization	[U]	Transport-Invariant Normalization [U]
thm:IV.AMCT	[U]	AMCT — Admissible Morphism Closure Theorem [U]
thm:IV.completion-ladder	[U]	Completion Ladder Theorem [U]

Full Spine Dependency Chain (Now Machine-Traceable)

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thm:GR.domain [C]
→ thm:VII.6.4 [U] (Closure Schema)
→ thm:V.UBLT [U] (UBLT Thm. V.8.1)
→ cor:IV.descendants-factor [U] (All realizations factor through U)
→ thm:IV.universal-completion [U] (Existence of U, Thm. IV.5.1)
→ def:IV.candidate-U [D] (U = T/~_pres)
→ def:IV.POT-category [D] (POT category)
→ def:IV.POT-object [D] (Sigma, Lambda, T, D quadruple)
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→ def:I.obs-quotient [D] ($\Sigma_n = \Omega/\sim_n$)
→ thm:I.stabilization [U] (Finite Stabilization)

Full Claim Catalog

Machine ID	Kind	St.	Title
sr:IV.source-not-branch	standing_r	[D]	Source-Not-Branch Standing Rule
def:IV.POT-object	definition	[D]	POT Object
def:IV.POT-category	definition	[D]	POT Category
def:IV.transport-invariant	definition	[D]	Transport Invariant
def:IV.candidate-U	definition	[D]	Candidate Universal Source U
def:IV.exhaustive-source-image	definition	[D]	Exhaustive Source-Image Condition
thm:IV.ORA	theorem	[U]	Overlap Route-Agreement Theorem (ORA)
thm:IV.CTM	theorem	[U]	Coherent Transition-Map Theorem (CTM)
thm:IV.normalization	theorem	[U]	Transport-Invariant Normalization Theorem
thm:IV.AMCT	theorem	[U]	Admissible Morphism Closure Theorem (AMCT)
thm:IV.completion-ladder	theorem	[U]	Completion Ladder Theorem
thm:IV.universal-completion	theorem	[U]	Universal Completion Theorem (Thm. IV.5.1)
thm:IV.source-uniqueness	theorem	[U]	Certified Uniqueness of the Universal Source
cor:IV.descendants-factor	corollary	[U]	All Descendants Factor Through U
thm:IV.terminal-source	theorem	[U]	Unified Terminal-Source Theorem
thm:IV.governing	theorem	[U]	Universal Source Theorem (Book IV Governing)
cor:IV.all-branches-through-U	corollary	[U]	Every Branch Book Attaches Through Book IV
thm:IV.unified-emergence	theorem	[C]	Unified Emergence Theorem (conditional)
status:IV	remark	[R]	Book IV Status — primarily unconditional; existence